

Macroeconomic Theory and Policy: Lecture 9 (Ch.6)

Experiment 2: An Increase in Research Share

- An Increase in Research Share, $\bar{\ell}$, has two effect to the growth rate
 - ▶ More researcher producing ideas

- ▶ Less labour producing output

$$y_t^* = \left(\frac{1}{\bar{n}}\right)^\gamma \bar{z}^\gamma (1 - \bar{\ell}) (\ell_{a,t})^\gamma = \left(\frac{1}{\bar{n}}\right)^\gamma \bar{z}^\gamma (1 - \bar{\ell}) (\ell N_t)^\gamma$$

$$g_y^* = \gamma \bar{n}$$

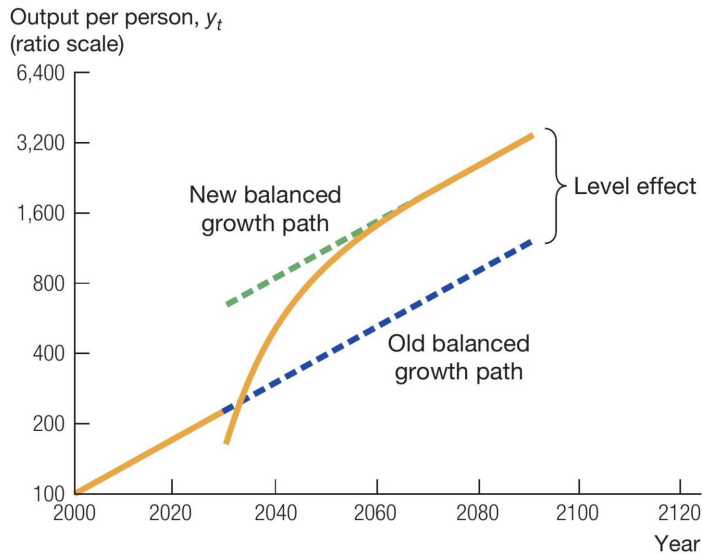
- In BGP, the growth rate would not change
- There are two opposite effect on the output per capita
- Increasing the research share thus involves a tradeoff: current output declines but more idea created which may increase the output in the long run

Experiment 2: An Increase in Research Share

In transition,

- The level of output per capita decreases initially
 - ▶ The number of worker in production sector drop
 - ▶ New ideas have not created
- From $t + 1$ onward, more idea will be created and increase the output as would in transition dynamic
- For total output, it could still increase if $\bar{n}$ is relatively high
 - ▶ Recall the total output function: $Y_t = A_t L_{y,t} = A_t(1 - \ell)N_t$
 - ▶ That is, if the positive effect from an increase in N_t is more than negative effect from an increase in ℓ
 - ▶ We have all the number to measure the changes, the results on total output is ambiguous

Experiment 2: An Increase in Research Share

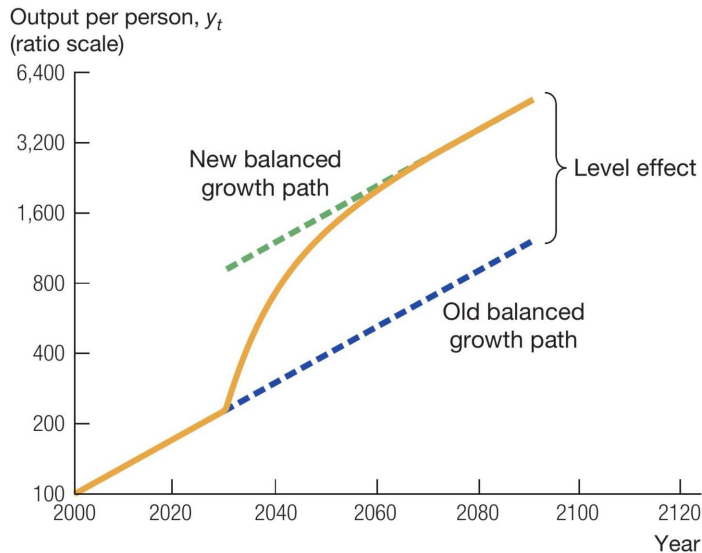


Experiment 3: A Permanent Rise in Research Productivity

$$y_t^* = \left(\frac{1}{\bar{n}}\right)^\gamma \bar{z}^\gamma (1 - \bar{\ell}) (L_{a,t})^\gamma = \left(\frac{1}{\bar{n}}\right)^\gamma \bar{z}^\gamma (1 - \bar{\ell}) (\ell N_t)^\gamma$$
$$g_y^* = \gamma \bar{n}$$

- In BGP, the growth rate would not change. But, the long-run level of output will be higher as If research is more productive
- In transition, there would be no change in the initial period since the A_t would not change until in period $t + 1$
- The A would grow faster but as the stock of A get larger, g_A would converge to the BGP

Experiment 3: A Permanent Rise in Research Productivity



Growth Accounting

- Growth accounting analyze the sources of growth in an economy and how they may change over time
- The approach is similar to development accounting but we will focus on growth
- Consider following production function

$$Y_t = A_t K_t^{1/3} L_{y,t}^{2/3}$$

- Here all variables can grow

Growth Accounting

$$Y_t = A_t K_t^{1/3} L_{y,t}^{2/3}$$

- Apply growth rate rules to the production function

$$g_{Y_t} = g_{A_t} + g(K_t^{1/3}) + g(L_{y,t}^{2/3})$$

$$g_{Y_t} = g_{A_t} + \frac{1}{3}g_{K_t} + \frac{2}{3}g_{L_{y,t}}$$

- The growth rate of output is the sum of the growth rate of each input weighted by its exponent

Growth Accounting

- We want to adjust growth rates by labour hours to account for structural change (e.g., rise in female labour participation) and business cycle
 - ▶ When the economy is booming, people may work more hours per week
 - ▶ When the economy is in a recession, people may work fewer hours per week
- In actual applications of growth accounting, the second term can include increases in education or changes in the age distribution of the workforce. In general, we call it “labour composition”

Growth Accounting

- Growth rate of output per hour, Y/L , over a time period is the sum of three terms:

$$g_{Yt} - g_{Lt} = \frac{1}{3}(g_{K_t} - g_{L_t}) + \frac{2}{3}(g_{L_{yt}} - g_{L_t}) + g_{At}$$

- Except for g_{At} , we can measure all the other terms. We can measure the unobserved TFP growth by backing out the term using our equation

Growth Accounting in the United States

Output per hour grew at an average annual rate of 2.3 percent between 1948 and 2017

- 0.9 percentage points were due to an increase in the capital labour ratio, K/L
- 0.2 percentage points came from changes in the composition of the labour force, including increased years of education
- TFP growth accounted for the majority of growth, at 1.2 percentage points

TABLE 6.2

Growth Accounting for the United States

	1948–2017	1948–1973	1973–1995	1995–2003	2003–2017
Output per hour, Y/L	2.3	3.3	1.5	3.2	1.4
Contribution of K/L	0.9	1.0	0.8	1.4	0.5
Contribution of labor composition	0.2	0.2	0.2	0.3	0.2
Contribution of TFP, A	1.2	2.1	0.5	1.5	0.7

The table shows the average annual growth rate (in percent) for different variables.

Source: Bureau of Labor Statistics, *Multifactor Productivity Trends*.

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Growth Accounting in the United States

- The period 1948 to 1973 featured the fastest growth in output per hour, at 3.3 percent per year
- The years 1973 to 1995 saw output per hour grow less than half as fast, at 1.6 percent per year
 - ▶ Known as the productivity slowdown
- In the years 1995–2003, output grew nearly as rapidly as before 1973–1995 (3.2%)
 - ▶ Known as the new economy
- In recent year, the growth has slow down again

Growth Accounting in the United States after 1995

- During this time was the emergence of the World Wide Web and the dot-com boom in the stock market, followed by the sharp decline in stock prices in 2001
- Growth accounting tells us this should be roughly equal parts by capital accumulation and TFP
- The contribution of capital rose from 0.8 percent to 1.4 percent
- TFP growth increased from 0.5 percent to 1.5 percent
- Economists studying this productivity boom generally conclude that at least half of it can be explained by purchases of computers and by rapid growth in the sectors that produce information technology
- That is, there is a link between the new economy and information technology

Combining Solow and Romer: Overview

- Solow endogenized capital accumulation and found that capital could not provide the engine of economic growth
- However, he went further in postulating that exogenous improvements in technology could explain growth
- The limitation of this approach, however, is that the growth was simply assumed rather than explained within the model
- Solow intuited that growth must be related to technological improvements, but he took these to be exogenous, like rain falling from the sky

Combining Solow and Romer: Overview

- Simplifying assumption in the Romer model that there was no capital
- But, it would be important to include capital accumulation into the model
- Nonrivalry of ideas results in long-run growth along a balanced growth path
- With capital, the model exhibits transition dynamics if economy is not on its balanced growth path in the Solow model sense
 - ▶ For short periods of time, countries can grow at different rates
 - ▶ In the long run, countries grow at the same rate if all parameters are the same

Combining Solow and Romer

- Compare to the Solow model, productivity level $\bar{A}$ is no longer a constant parameter
 - ▶ Now, the $\bar{A}$ is replaced by A_t , which grow continuously
- In a Solow diagram, this would show up as the $\bar{s}y_t = \bar{s}A_t k_t^\alpha$ curve shifting upward each period, leading the capital stock to increase each period as new investment exceeded depreciation

Setting Up the Combined Model

- Six equations:

$$Y_t = A_t K_t^\alpha L_{yt}^{1-\alpha}$$

$$\Delta K_{t+1} = \bar{s} Y_t - \delta K_t$$

$$\Delta A_{t+1} = \bar{z} L_{at}$$

$$L_{yt} + L_{at} = N_t$$

$$L_{at} = \bar{\ell} \bar{L}$$

$$\frac{\Delta N_{t+1}}{N_t} = \bar{n}$$

- we have 6 unknowns (Y_t , K_t , A_t , L_{yt} , N_t and L_{at}) in each period t

Setting Up the Combined Model

- Convert the per capita production function ($y_t = A_t^\gamma k_t^\alpha (1 - \ell)^{1-\alpha}$) into growth rate

$$g_{yt} = \gamma g_{At} + \alpha g_{kt}$$

- The change in the capital per capita is investment minus depreciation and population growth:

$$g_{kt} = \frac{\Delta k_{t+1}}{k_t} = \bar{s} \frac{y_t}{k_t} - \delta - \bar{n}$$

- Researchers are used to produce new ideas:

$$g_{At} = \frac{\Delta A_{t+1}}{A_t} = \bar{n}$$

Setting Up the Combined Model

- This equation still has two endogenous variables on the right side, so it's not yet a solution

$$g_{kt} = \frac{\Delta k_{t+1}}{k_t} = \bar{s} \frac{y_t}{k_t} - \delta - \bar{n}$$

- For g_{kt} to grow at constant rate, it must be the case that the right hand side is also a constant
- Given $\bar{n}$, $\bar{s}$ and δ is constant, $\frac{y_t}{k_t}$ must be constant
- It can only be possible if y_t and k_t is growing at the same rate
 - ▶ If y_t or k_t is growing at a faster rate, each period the ratio $\frac{y_t}{k_t}$ would grow bigger or smaller
- Therefore, $g_k^* = g_y^*$ along the balanced growth path

Setting Up the Combined Model

- Plug in $g_{At} = \bar{n}$, $g_k^* = g_y^*$ into $g_{yt} = \gamma g_{At} + \alpha g_{kt}$,

$$g_y^* = \gamma \bar{n} + \alpha g_y^*$$

- Rearrange it, we have

$$g_y^* = \frac{\gamma \bar{n}}{1 - \alpha}$$

- For the long-run combined model, this equation pins down the growth rate of output per person

Setting Up the Combined Model

- The growth rate of output is even larger in the combined model than in the Romer model (the extra term, $\frac{1}{1-\alpha} > 1$)
- Output is higher in this model because ideas have a direct and an indirect effect
 - ▶ Increasing productivity raises output because productivity has increased
 - ▶ Higher productivity results in a higher capital stock
- While capital cannot itself serve as an engine of long run economic growth, it helps to amplify the underlying growth in knowledge

Setting Up the Combined Model

- Now, we know that g_Y^* is a constant along a balanced growth path (It would have an extra of $(1 - \alpha)\bar{n}$). This implies

$$\frac{k_t^*}{y_t^*} = \frac{K_t^*}{Y_t^*} = \frac{\bar{s}}{g_Y^* + \delta}$$

- Substitute $k^* = \frac{\bar{s}}{g_Y^* + \delta} y^*$ into the production function:

$$\begin{aligned} y_t^* &= A^\gamma \left(\frac{\bar{s}}{g_Y^* + \delta} y^* \right)^\alpha (1 - \ell)^{1-\alpha} \\ \implies y_t^{*(1-\alpha)} &= A^\gamma \left(\frac{\bar{s}}{g_Y^* + \delta} \right)^\alpha (1 - \ell)^{1-\alpha} \\ \implies y^* &= A_t^{*\frac{\gamma}{1-\alpha}} \left(\frac{\bar{s}}{g_Y^* + \delta} \right)^{\frac{\alpha}{1-\alpha}} (1 - \bar{\ell}) \\ \implies y^* &= A_t^{*\frac{\gamma}{1-\alpha}} \left(\frac{\bar{s}}{g_Y^* + \delta + \bar{n}} \right)^{\frac{\alpha}{1-\alpha}} (1 - \bar{\ell}) \end{aligned}$$

- Here, $\bar{s}$ would lead to higher y as in Solow model

Combined Model - Transition Dynamic

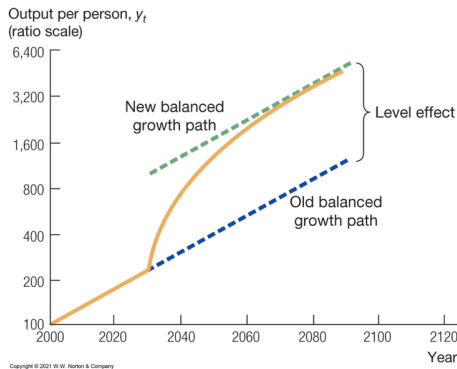
- The Solow model and the combined model both have diminishing returns to capital
- Transition dynamics applies in both models
- In the combined model,
 - ▶ the further below its balanced growth path an economy is, the faster the economy will grow
 - ▶ the further above its balanced growth path an economy is, the slower the economy will grow
- Note that the K/Y is not equal to $\frac{\bar{s}}{g_Y^* + \delta}$ when it is not on the balanced growth path

Combined Model - Transition Dynamic

- A permanent increase in the investment rate $\bar{s}$ in the combined model implies that
 - ▶ The balanced growth path of income is higher (parallel shift)
 - ▶ Current income is unchanged, which implies The economy is now below the new balanced growth path
 - ▶ The growth rate of income per capita is immediately higher (transition dynamic)
 - ▶ The slope of the output path is steeper than the balanced growth path and get flatter overtime until it reach the balance growth path

Combined Model - Transition Dynamic

- The economy begins on a balanced growth path. Then, in the year 2030, there is a permanent increase in the investment rate, s , which raises the balanced growth path.
- Since the economy is now below its new balanced growth path, the principle of transition dynamics says that the economy will grow rapidly.
- Notice that y_t is plotted on a ratio scale, so the slope of the path is the growth rate



Combined Model - Transition Dynamic

- The principle of transition dynamics played a crucial role in explaining variations in growth rates among countries in the Solow model in the short run
- The combined model presents a theory of long-run growth driven by the global discovery of new ideas, alongside a theory explaining differences in growth rates among countries based on transition dynamics
- The model predicts that, in the long run, all countries should grow at the same rate, denoted by $\bar{g}$, representing the world knowledge growth rate,
- However, within any specific time frame, differences in growth rates among countries may be observed due to varying distances from their balanced growth paths
- Alterations in policies impacting model parameters, such as the investment rate, can consequently lead to divergent growth rates over extended periods